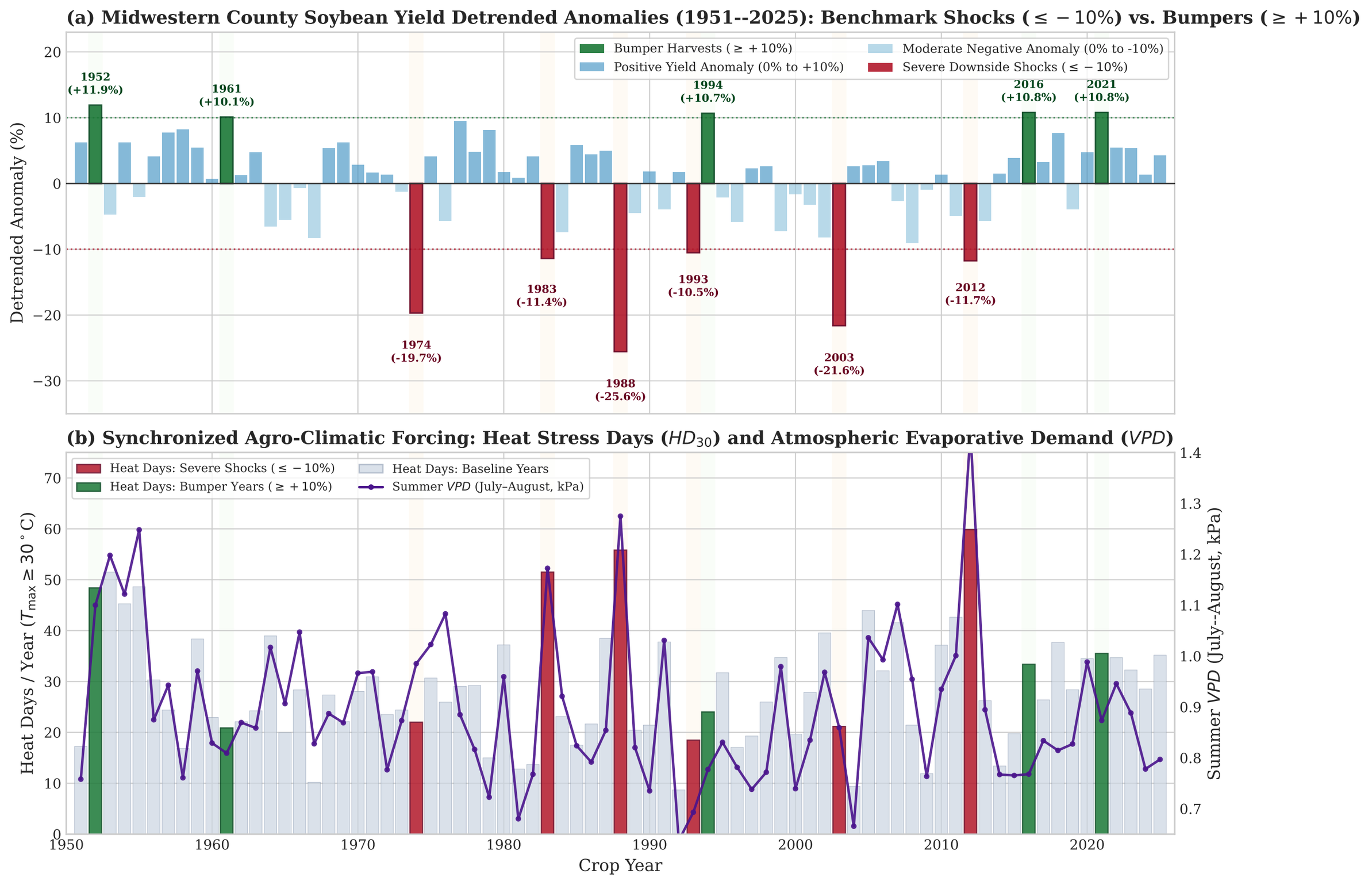

(a) Midwestern County Soybean Yield Detrended Anomalies (1951--2025): Benchmark Shocks (≤ −10%) vs. Bumpers (≥ +10%)
Bumper Harvests (≥ +10%)
Positive Yield Anomaly (0% to +10%)
Moderate Negative Anomaly (0% to -10%)
Severe Downside Shocks (≤ −10%)
Detrended Anomaly (%)
1952 (+11.9%)
1961 (+10.1%)
1974 (-19.7%)
1983 (-11.4%)
1988 (-25.6%)
1993 (-10.5%)
1994 (+10.7%)
2003 (-21.6%)
2012 (-11.7%)
2016 (+10.8%)
2021 (+10.8%)
(b) Synchronized Agro-Climatic Forcing: Heat Stress Days (HD_{30}) and Atmospheric Evaporative Demand (*VPD*)
Heat Days: Severe Shocks (≤ −10%)
Heat Days: Bumper Years (≥ +10%)
Heat Days: Baseline Years
Summer *VPD* (July–August, kPa)
Heat Days / Year (T_{max} ≥ 30 °C)
Summer *VPD* (July--August, kPa)
Crop Year